

Experiment-3 write the python program for water jug problem

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water jug.py - C:/Users/prana/OneDrive/Desktop/water jug.py (3.13.14)
File Edit Format Run Options Window Help

a = 0
b = 0

print("Initial State: (0, 0)")

b = 3
print("Fill Jug 2:", a, b)

a = b
b = 0
print("Pour Jug 2 into Jug 1:", a, b)

b = 3
print("Fill Jug 2:", a, b)

b = b - (5 - a)
a = 5
print("Pour Jug 2 into Jug 1:", a, b)

a = 0
print("Empty Jug 1:", a, b)

a = b
b = 0
print("Pour Jug 2 into Jug 1:", a, b)

if a == 1:
    print("Goal achieved!")
```

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IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/prana/OneDrive/Desktop/water jug.py =====
Initial State: (0, 0)
Fill Jug 2: 0 3
Pour Jug 2 into Jug 1: 3 0
Fill Jug 2: 3 3
Pour Jug 2 into Jug 1: 5 1
Empty Jug 1: 0 1
Pour Jug 2 into Jug 1: 1 0
Goal achieved!
>>>
```